International Capital Flows in the Context of Financial Stability

1MT404 Financial Stability

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Outline

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Balance of Payments (BoP) Framework (BPM6)

The fundamental identity

The BoP records every transaction between residents and non-residents twice (double entry):

$$CA + KA + NEO = FA$$

$FA =$ net acquisition of foreign assets minus net incurrence of foreign liabilities (net lending to the rest of the world); $NEO =$ net errors and omissions.

A current account deficit is not "lost trade": it is the counterpart of a net capital inflow – a net increase in liabilities to non-residents.

- **Current account (CA)**: goods, services, primary income (investment returns), secondary income
- **Capital account (KA)**: capital transfers, non-produced non-financial assets – small, except EU investment grants in Central Europe
- **Financial account (FA)**: FDI, portfolio investment, other investment (loans, deposits), reserve assets

FA is a flow; the *stock* of net foreign assets also moves with prices and exchange rates (valuation effects, NIIP slides).

Sectoral Balances and the Macroeconomic Identity

From the national income identity $Y = C + I + G + (X - M)$ we derive the sectoral financial balances:

$$(S_p - I_p) + (T - G) = (X - M) \approx CA$$

- $(S_p - I_p)$: **private sector balance**
- $(T - G)$: **public sector balance** (government surplus or deficit)
- $(X - M)$: **foreign sector balance**; the current account adds net primary and secondary income

Core implication: a current account deficit ($CA < 0$) *must* be matched by a private sector deficit (e.g. US household borrowing before 2008), a public sector deficit, or both. It is a saving–investment gap, not just a trade phenomenon.

The Kalecki Profit Equation: An Accounting Lens

Assuming workers consume what they earn ($C_w = W$), aggregate profits (P) are

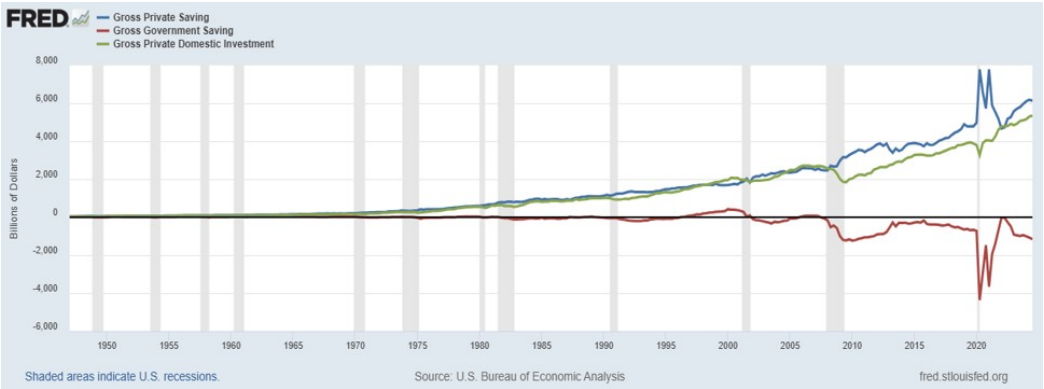
$$P = I + (G - T) + (X - M) + C_k - S_w$$

Profits = investment + government deficit + current account surplus + capitalist consumption − workers' saving.

- **Export-led economies (Germany, China):** the $(X - M)$ term supports profits without domestic public or private deficits
- **Deficit economies (US, UK):** with $(X - M) < 0$, profits are supported by government deficits $(G - T)$ or by debt-financed investment and consumption

An identity, not a causal model: it shows where profits come from in the accounts, not what drives what (Kalecki; Lavoie 2014).

US Gross Private Saving, Investment and Government Saving



Takeaway: Since 2008 US private saving has exceeded private investment, and the gap mirrors the government deficit – the sectoral identity at work.

Source: FRED (gross private saving, gross government saving, gross private domestic investment; USD billion, quarterly).

The Intertemporal Approach: Core Theory

The intertemporal approach (IA) views the current account as the optimal, forward-looking outcome of consumption-smoothing and investment decisions (Obstfeld and Rogoff 1996).

Intertemporal budget constraint (IBC): with B_t = net foreign *assets* at the end of period t and TB = trade balance,

$$(1+r)B_{t-1} + \sum_{s=t}^{\infty} \frac{TB_s}{(1+r)^{s-t}} = 0$$

A debtor ($B_{t-1} < 0$) must run future trade surpluses whose present value repays its net foreign debt, $-(1+r)B_{t-1}$ – no Ponzi schemes.

Consumption smoothing:

- **Temporary shocks:** a temporary fall in income (e.g. a natural disaster) leads to a CA deficit as the country borrows to smooth consumption
- **Permanent shocks:** a permanent fall in income leads to an immediate cut in consumption, leaving the CA largely unchanged

Two-Period Small Open Economy Model

Two periods, one good, exogenous world rate r, endowments Y_1, Y_2, $U = u(C_1) + \beta u(C_2)$; $B_1 =$ net foreign assets after period 1, with $B_0 = B_2 = 0$.

Budget constraints:

$$C_1 + B_1 = Y_1, \qquad C_2 = Y_2 + (1+r)B_1$$

Intertemporal budget constraint:

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$$

Utility maximisation $\Rightarrow$ Euler equation:

$$u'(C_1) = \beta(1+r)\,u'(C_2)$$

If $\beta(1+r) = 1$, consumption is flat (any concave u):

$$C_1 = C_2 = \frac{(1+r)Y_1 + Y_2}{2+r}$$

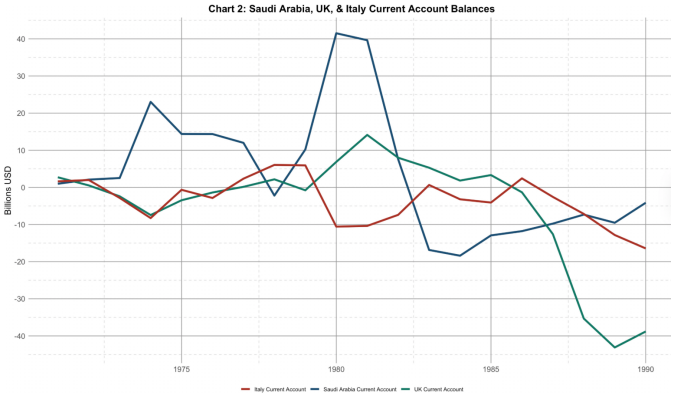
$$CA_1 = Y_1 - C_1 = B_1$$

A surplus when Y_1 is temporarily high; a deficit when Y_1 is temporarily low.

Applications: Oil Shocks, Immigration and Debt Crises

- **1970s oil shocks:** exporters (Saudi Arabia) ran large surpluses and lent them abroad; importers borrowed to smooth consumption (next slide)
- **Israel, 1990s immigration wave:** a current account deficit driven by *investment* – housing and capital for about a million immigrants – the IA's investment channel
- **US–China imbalances:** the US deficit reflects low saving. China's surplus is a *puzzle* for the IA: a fast-growing economy should borrow against its future income, not lend – the "allocation puzzle" (Gourinchas and Jeanne 2013)
- **Sovereign debt:** debt is sustainable only if the future surpluses it requires are feasible; Greece needed surpluses but ran deficits

Intertemporal Approach Example: The Oil Shocks of the 1970s



Takeaway: Saudi Arabia saved its windfalls (surpluses of about \$23 bn in 1974 and \$41 bn in 1980) and borrowed once oil prices fell; Italy borrowed after the shocks – smoothing in practice.

Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.

Current Account Surpluses in Developed Economies

Structural ("natural") drivers:

- Ageing populations $\Rightarrow$ high private saving
- A positive net international investment position ("rentier" status) $\Rightarrow$ net investment income
- A high capital/labour ratio $\Rightarrow$ diminishing returns to domestic investment
- Technological leadership and high value-added exports
- Low investment relative to saving

These create persistent surpluses without policy intervention – although policy (wage restraint, fiscal rules, exchange rate regimes) can reinforce them.

The Lucas Paradox and "Uphill" Capital Flows

The paradox (Lucas 1990): capital should flow to capital-scarce poor countries, where returns should be higher – but far too little does. **Strong form:** net flows go "uphill", from poorer to richer countries (Prasad, Rajan and Subramanian 2007).

Why?

- Lower risk-adjusted returns: weaker institutions, human capital, sovereign risk (Alfaro, Kalemli-Ozcan and Volosovych 2008)
- Official flows: reserve accumulation by EM central banks drives the uphill flow; private capital still flows downhill (Alfaro et al. 2014)
- Faster-growing EMs receive *less* capital (Gourinchas and Jeanne 2013)

On the receiving side – why rich deficit countries borrow:

- Falling household saving; consumption growth above productivity growth
- Low real rates and rising household and public debt
- Booms in non-productive investment (housing, real estate)
- Rising unit labour costs and lost competitiveness

From Theory to Crisis: What the IA Misses

So far: the IA explains why countries run surpluses or deficits as rational, forward-looking behaviour.

The problem: it models *net* resource transfers. Persistent current account deficits signal a **medium- to long-term vulnerability** (structural fragility), but they are rarely the immediate cause of a crisis.

Modern financial crises are almost always triggered by **sudden stops in gross financial flows** – bank lending, wholesale funding, portfolio flows – dynamics that the current account balance does not capture.

Two Views of Global Imbalances Before 2008

	Global saving glut (Bernanke 2005)	**Global banking glut / gross flows** (Bernanke et al. 2011; Shin 2012; Borio and Disyatat 2011)
Story	Excess saving in Asia and oil exporters flows to the US, lowers long-term rates and fuels the housing boom	European banks borrow dollars in the US and buy US mortgage securities; gross flows, not net, finance the boom
Key variable	Current account (net flows)	Gross positions, funding and leverage
Evidence	Large Asian surpluses and reserve accumulation; low long-term yields (the "conundrum")	The euro area's current account was close to balance, yet European investors bought a large share of US private-label mortgage securities
Policy	Rebalance saving and demand across countries	Monitor and regulate gross exposures, dollar funding and bank leverage

Takeaway: Both views capture part of the story – but only the gross-flow view explains why European banks, not Asian central banks, needed dollar support in 2008.

The Breakdown of the Intertemporal Approach

While the IA models **real resource transfers** (net flows) well, it misses the mechanics of modern financial crises.

Critique by Borio, Disyatat and Shin:

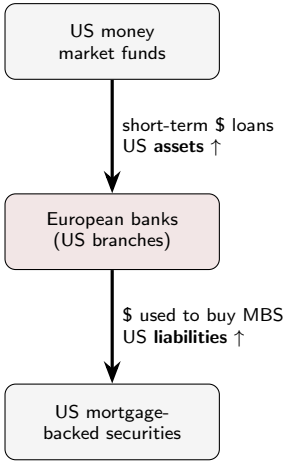
- **Saving is not financing:** the IA conflates saving (an ex-post national accounting residual – income not consumed) with financing (an ex-ante cash-flow concept – access to purchasing power)
- **Endogenous money:** banks do not merely intermediate pre-existing saving (the loanable funds view); they create purchasing power when they lend. Investment can be financed before saving exists, which decouples credit booms from the current account
- **The "excess elasticity" of credit:** fragility stems from the financial system's ability to expand balance sheets without a firm anchor in saving – not from a global saving glut

Gross vs. Net Capital Flows: The 2008 Example

Before 2008 the euro area's current account was close to balance, but **gross flows** with the US exploded:

- US money market funds lent dollars to European banks $\Rightarrow$ **US foreign assets** rise (claims on European banks)
- European banks used the dollars to buy US mortgage-backed securities $\Rightarrow$ **US foreign liabilities** rise
- Net flow ≈ 0; gross positions large and maturity-mismatched (short-term funding, long-term assets)

When mortgage values fell in 2007–08, the money funds pulled back: a **sudden stop** in dollar funding for European banks, contained only by Fed swap lines (FX lecture, slide 81).



net flow ≈ 0, gross flows large

Gross Flows: Surges, Stops, Flight and Retrenchment

Gross **inflows** (by foreigners)	**Surge** sharp rise	**Stop** sharp fall
Gross **outflows** (by residents)	**Flight** sharp rise	**Retrenchment** sharp fall

Forbes and Warnock (2012)

- Episodes are driven mainly by **global factors** – risk aversion, global growth – and spread through contagion; domestic factors matter less
- **Sudden stop** (Calvo 1998): an abrupt fall in net inflows forces the current account to adjust – a crisis when debt is short-term and in foreign currency

What a central bank monitors

Gross inflows and outflows by type and investor; short-term external debt relative to reserves; currency mismatches; non-resident holdings of local-currency bonds; rollover of bank funding.

What the Data Tell Us: A Road Map

Four empirical regularities to keep in mind

What we are looking for

Theory predicts that capital flows to where returns are highest. The data show a more complex picture.

1. **FDI is the anchor.** It is the least volatile component of capital flows and the least linked to sudden stops.
2. **Global imbalances are persistent.** The US has run a current account deficit almost every year since the early 1980s; East Asia has run surpluses.
3. **Saving rates explain the surplus side.** East Asia & Pacific saves about 35% of GDP (China over 40%); North America under 20%.
4. **Valuation effects matter as much as flows.** The US NIIP moves with the dollar and equity prices: between end-2024 and mid-2026 deficits of about \$1.6 trillion moved it by only \$0.3 trillion.

Empirical Patterns: FDI Inflows (1990–2023)

By development level

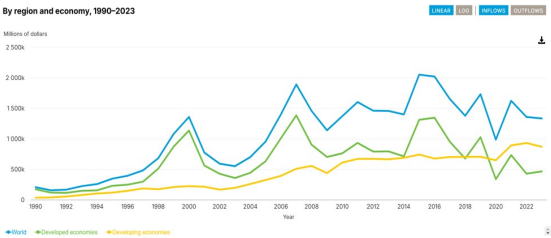
FDI went mostly to developed economies until the mid-2000s; flows to developing economies (above all Asia) have grown steadily since.

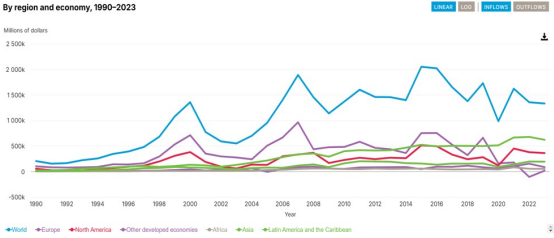
By region

- Asia: strong growth (China, then India and South-East Asia)
- Europe and North America: the largest but most volatile recipients
- Africa, Latin America and the Caribbean: smaller, commodity-driven

FDI is the *least volatile* component of capital flows. Portfolio and "other" flows (bank and wholesale funding) dominate volatility and crisis transmission.

FDI Inflows, 1990–2023

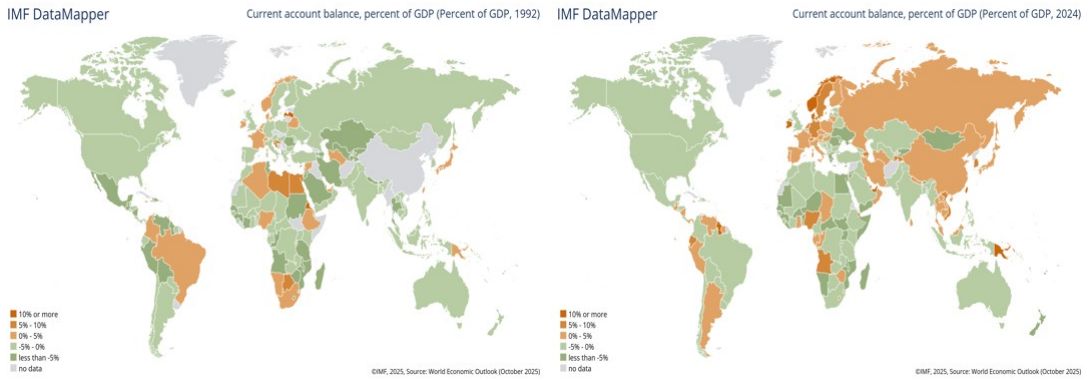




Takeaway: World FDI peaked in 2007 and 2015–16 and has been weaker since; flows to developing economies (left, yellow) have been the steadiest.

Source: UNCTAD, World Investment Report 2024 / UNCTADstat; FDI inflows, USD million; by development level (left) and by region (right).

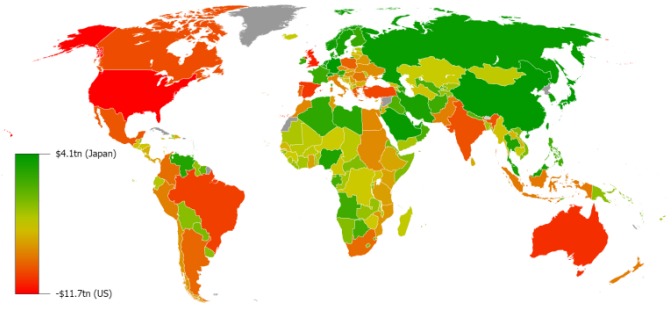
Current Account Balances, 1992 and 2024



Takeaway: By 2024 surpluses (orange) are concentrated in China, northern Europe and commodity exporters, while the US and most of Africa and Latin America run deficits (green).

Source: IMF, World Economic Outlook (October 2025), via IMF DataMapper; current account balance, % of GDP.

Cumulative Current Account Balances



Takeaway: Over the period shown, the US cumulative deficit reached about \$11.7 trillion and Japan's cumulative surplus about \$4.1 trillion – the two ends of the colour scale.

Source: D. Wales, lecture notes.

The Persistent US Current Account Deficit

- A deficit almost every year since the early 1980s, widest in 2006 (about 6% of GDP); in Q2 2026 $246 bn, or 3.0% of GDP (BEA)
- Financed by foreign purchases of US assets: as issuer of the main reserve currency, the US borrows in its own currency and needs no reserves
- Counterparts: low household and public saving, booms in non-tradables, asset price booms partly fuelled by inflows, a shrinking manufacturing export base

Deficits reflect saving–investment gaps, not "unfair trade" – although trade policy can shift where the deficit shows up bilaterally (next slides).

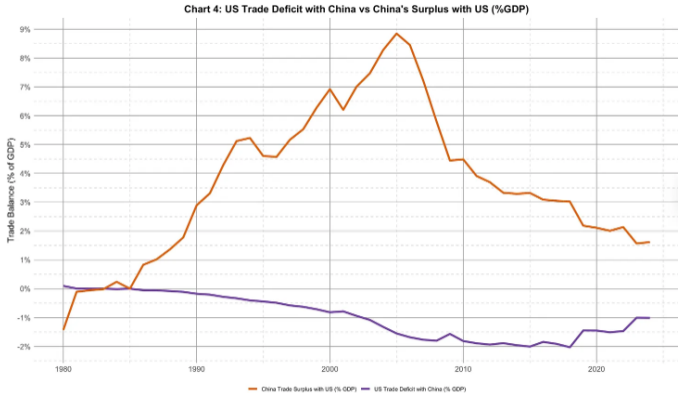
US and China: Current Account and Trade Balances



Takeaway: China's current account surplus peaked near 10% of GDP in 2007 and has been 1–3% since; the US deficit peaked near 6% in 2006.

Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.

US–China Bilateral Trade Balance



Takeaway: Measured against China's GDP, its surplus with the US fell from about 9% (2006) to under 2% (2024); against US GDP the bilateral deficit is now about 1%.

Source: Benigno, G. (2025), "Background note on trade and trade imbalances", Substack.

Asian Economies Financing Global Deficits

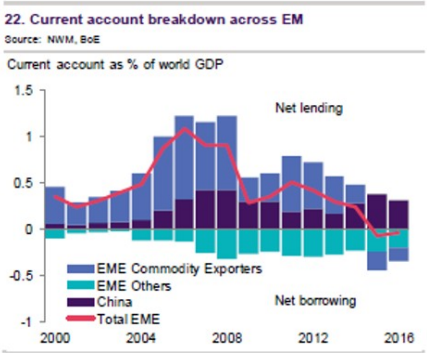
China and Asian emerging markets:

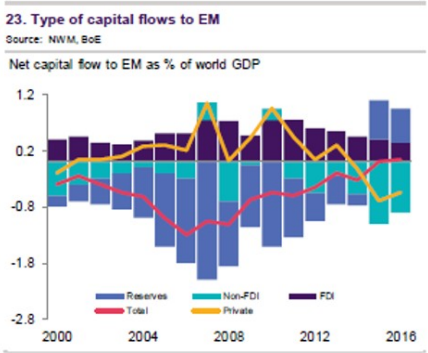
- High private saving and export-led growth (productivity surges)
- Massive reserve accumulation (self-insurance after 1997, and competitiveness through managed exchange rates)
- *Public* (not private) financing of developed-world deficits
- Young demographics initially boosted saving

Consequences:

- US debt was financed largely by EM official reserves – the "uphill" flow of the Lucas-paradox slide
- Global asset ownership shifted toward EM central banks and sovereign wealth funds

Emerging Markets: Current Accounts and Types of Capital Flows

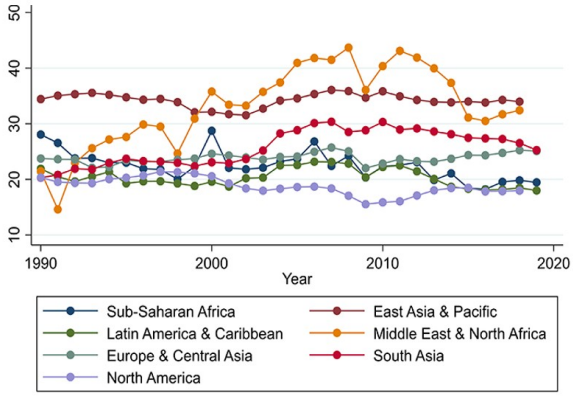




Takeaway: Before 2008 EM surpluses (mostly China's) were recycled as reserve accumulation – official outflows (blue, right) – while FDI and private inflows continued.

Source: NatWest Markets; Bank of England.

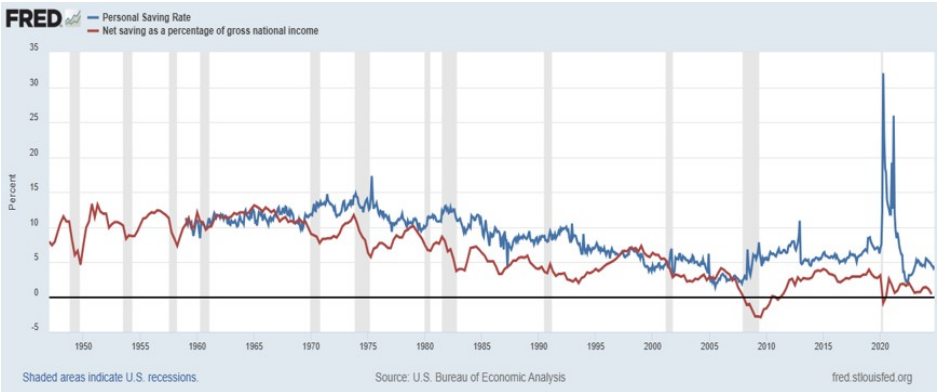
Saving Rates by Region



Takeaway: East Asia & Pacific saves about 35% of GDP and South Asia 25–30%, against under 20% in North America; MENA swings with oil prices.

Source: World Bank, World Development Indicators, gross savings (% of GDP), regional aggregates.

US Saving Rates

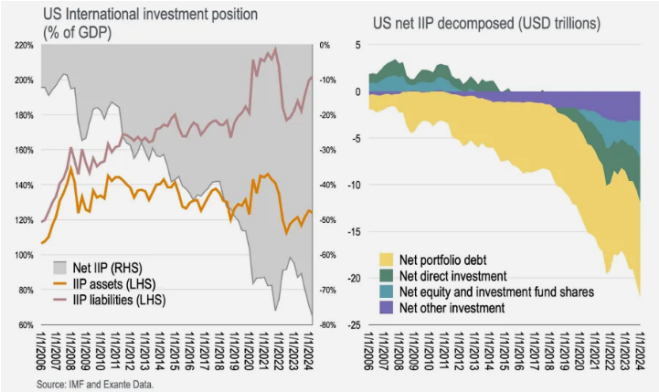


Takeaway: US net national saving fell from about 10% of GNI in the 1960s to around zero by 2008 and has stayed low; the 2020 spike in household saving was temporary.

Source: FRED (personal saving rate; net saving as % of gross national income).

US International Investment Position

Exorbitant privilege?

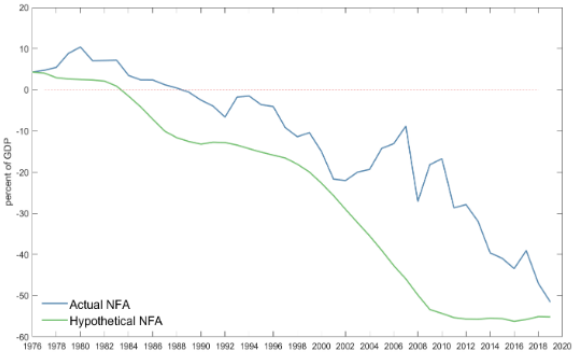


Takeaway: The US net position fell from about -10% of GDP (2007) to about -80% (2024), mostly because foreign holdings of US debt and equity securities grew.

Source: Exante Data (J. Nordvig), Money Inside Out (Substack), "United States: changing net IIP"; data: IMF.

The US NIIP and Valuation Changes

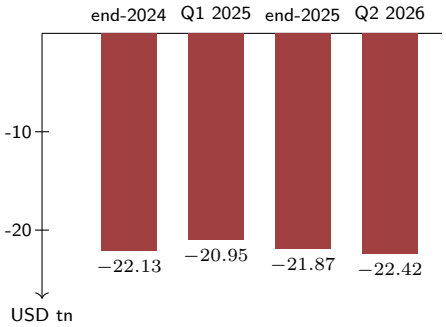
Actual vs. hypothetical NIIP without valuation changes since 1976



Takeaway: Until about 2010 valuation gains kept the US position far less negative than cumulated deficits imply; by 2019, when this chart ends, the gap had closed.

Source: G. Hall, Brandeis University, Econ 303 notes, "The NIIP paradox"; data: BEA.

2024–2026: Valuation at Work Again



US net international investment position, end of period (revised data)

- Current account deficits: $1.12 trillion in 2025 and about $0.46 trillion in the first half of 2026 – about $1.6 trillion in total
- Yet the NIIP fell by only about $0.3 trillion
- The difference, about $1.3 trillion, is mostly valuation: the weaker dollar in 2025 raised the dollar value of US-owned foreign assets

Takeaway: The NIIP is not cumulated deficits: over 18 months valuation offset about four-fifths of the deficits.

Source: U.S. Bureau of Economic Analysis.

NIIP: Beyond Trade Balances

The net international investment position (NIIP) is the value of US-owned foreign assets minus foreign-owned US assets.

$$\Delta NIIP_t = CA_t + \Delta VA_t + RES_t$$

where CA is the current account, ΔVA the **valuation changes** and RES other changes in volume.

- **Price effects:** if US stock markets outperform foreign markets, US *liabilities* rise faster than assets and the NIIP falls
- **Exchange rate effects ("exorbitant privilege"):** US liabilities are mostly in dollars, US foreign assets largely in foreign currencies – a **weaker dollar** raises the dollar value of the assets and mechanically improves the NIIP (Gourinchas and Rey 2007)

Takeaway: In 2020–24 US equity outperformance made the NIIP more negative; in 2025 the weaker dollar reversed part of it.

From Global Patterns to a Case Study

Pattern established: gross flows dominate volatility; FDI is stable, portfolio and banking flows are not.

Next: the euro area (1999–2012) is the clearest case where a sudden stop in gross banking flows – acting on the medium-term vulnerability of divergent unit labour costs – triggered a systemic crisis.

The Eurozone Debt Crisis: Gross Flows and the Real Exchange Rate

The euro area is a textbook case of capital flow misallocation within an incomplete optimum currency area (OCA).

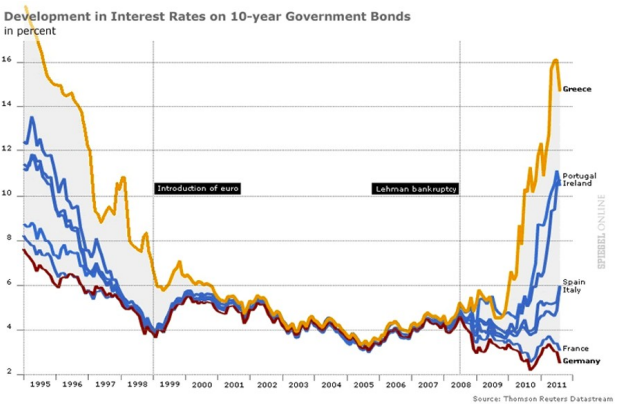
The setup (1999–2007):

- Monetary union removed nominal exchange rate risk and compressed sovereign spreads:

$$i^{\text{Periphery}} \downarrow \rightarrow \; i^{\text{Periphery}} \approx i^{\text{Core}}$$

- Large **gross capital flows** from the core (e.g. Germany, France) to the periphery (e.g. Spain, Greece), intermediated by banks
- The inflows financed domestic demand booms – Spain: housing and construction; Greece: public consumption and fiscal expansion
- Weak allocation $\rightarrow$ resources moved into non-tradables

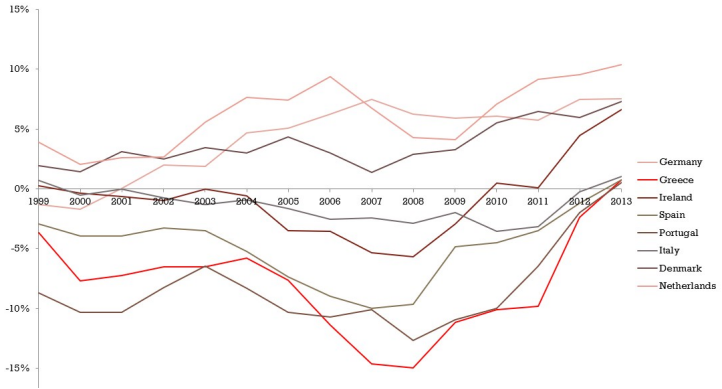
Ten-Year Government Bond Yields, 1995–2011



Takeaway: Spreads vanished after the euro's introduction – markets priced Greek and German debt almost alike – and reopened only after 2008.

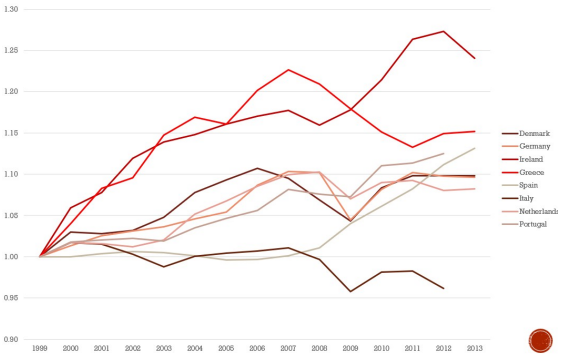
Source: Spiegel Online; data: Thomson Reuters Datastream.

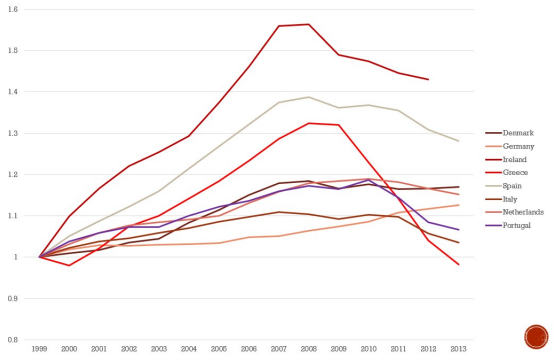
Current Account Balances, 1999–2013



Takeaway: Greece, Portugal and Spain ran deficits of 10–15% of GDP by 2008 while Germany and the Netherlands moved into large surpluses; the deficits closed only through the 2010–13 contraction.

Productivity and Consumption, 1999 = 1





Takeaway: Demand ran ahead of productivity in Ireland, Spain and Greece: consumption (right) rose 30–55% between 1999 and 2008, then fell back after 2010.

Unit Labour Costs (ULC) and Cost Pressures

Nominal unit labour cost (w = nominal wage, L = labour, Y = real output):

$$ULC_n = \frac{wL}{Y} = \frac{w}{Y/L}$$

- ULC_n rises when nominal wage growth (Δw) exceeds productivity growth (Δa):

$$\Delta ulc_n \approx \Delta w - \Delta a$$

- **Periphery:** large capital inflows pushed up wages without matching productivity gains: $\Delta w \gg \Delta a \Rightarrow ULC_n \uparrow$
- **Core (e.g. Germany):** wage moderation kept costs in check: $\Delta w \approx \Delta a \Rightarrow ULC_n$ stable

Real Unit Labour Costs and the Labour Share

Real ULC (ULC_r) deflates nominal ULC by the price level (P):

$$ULC_r = \frac{ULC_n}{P} = \frac{wL}{PY} = \frac{w/P}{Y/L}$$

- Real ULC is the ratio of the **real wage** (w/P) to **labour productivity** (Y/L)
- Economically, it is the **labour share of income** – the part of national income paid to workers
- In growth rates: $\Delta ulc_r \approx \Delta w_{real} - \Delta a$

For the euro crisis: nominal ULC measured the loss of export competitiveness; real ULC tracked whether workers or capital owners captured the gains of the boom.

The Real Exchange Rate Channel

With fixed nominal exchange rates inside the euro area, cost competitiveness depends on relative nominal ULCs:

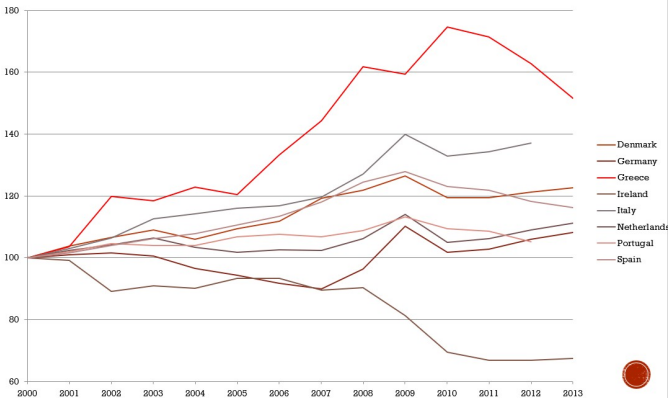
$$REER \sim \frac{ULC^{\text{Periphery}}}{ULC^{\text{Core}}}$$

- Rising relative ULC $\rightarrow$ **real appreciation** $\rightarrow$ lost competitiveness (exports $\downarrow$, imports $\uparrow$) $\rightarrow$ current account deficits financed by gross inflows

Crisis dynamics (2010 onwards):

- A sudden stop in gross private flows $\rightarrow$ funding crisis in the banking systems
- No nominal devaluation possible (monetary union)
- Adjustment through **internal devaluation**: wage compression, high unemployment and severe contraction

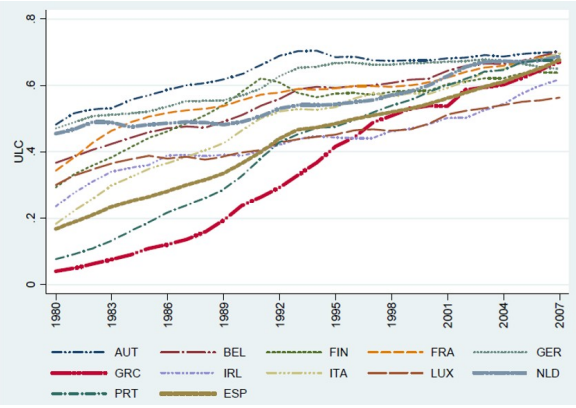
ULC Divergence: Core vs. Periphery (Nominal)



Read-off:

- Greece's nominal ULC: about $+60\%$ by 2008 and $+75\%$ by 2010 ($2000 = 100$); Germany flat or falling
- Mechanism: capital inflows $\Rightarrow$ wage growth $\gg$ productivity growth
- No devaluation possible $\Rightarrow$ competitiveness loss baked in

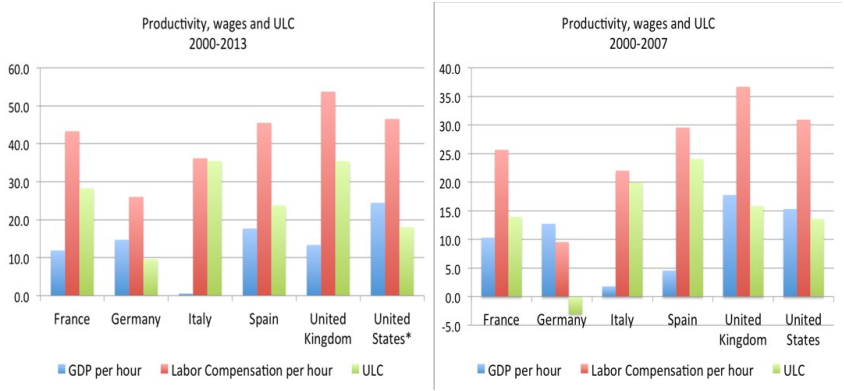
Unit Labour Costs over the Longer Run, 1980–2007



Takeaway: Over 1980–2007 ULCs rose fastest in Greece, Portugal and Spain – from low starting levels – converging on the core by the late 2000s.

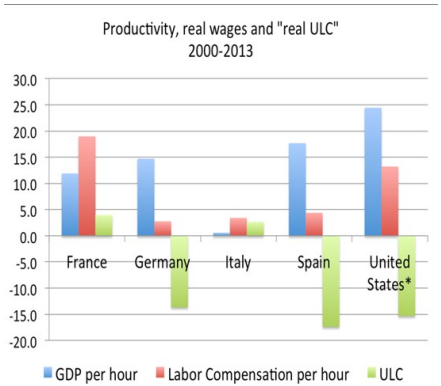
Source: Felipe, J. and Kumar, U. (2011), "Unit labor costs in the Eurozone: the competitiveness debate again", Levy Economics Institute WP 651; data: OECD.

Productivity, Wages and ULC, 2000–2013 and 2000–2007



Takeaway: In 2000–07 German compensation per hour rose less than productivity, so its ULC fell; in Spain and Italy compensation outran productivity by about 20 percentage points.

Productivity, Real Wages and Real ULC, 2000–2013



Takeaway: Deflated by prices, real ULC fell by about 14% in Germany and 17% in Spain: the labour share fell even where nominal ULC rose.

The "Competitiveness Paradox" (2000–2007)

In the periphery's boom years, **nominal ULC surged while real ULC stayed flat or fell.**

The mechanism:

- Capital inflows drove domestic demand $\rightarrow$ high non-tradable inflation ($P \uparrow$)
- Tight labour markets pushed up nominal wages ($\Delta w > 0$)
- With stagnant productivity ($\Delta a \approx 0$), **nominal ULC surged**

Why real ULC fell ($ULC_r = ULC_n / P$):

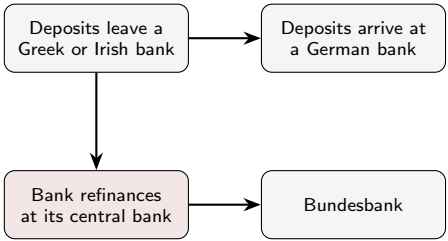
- Firms had pricing power and raised prices faster than wages, so P grew faster than ULC_n

The policy error:

- Real ULC (the labour share) suggested labour was getting "cheaper" relative to output
- But rising nominal ULC was pricing these countries out of export markets and building external imbalances

TARGET2: When Private Money Left, the Eurosystem Stepped In

- When private lenders and depositors pulled out of periphery banks, the banks refinanced at their national central bank
- The payments ran through TARGET2: the periphery central bank's **liability** to the Eurosystem rose; the Bundesbank's **claim** rose
- April 2012: periphery central banks' TARGET2 liabilities **€851 bn**; Bundesbank claims **€644 bn** (Lubik and Rhodes 2012)
- In Ireland the rise in TARGET2 liabilities was a multiple of the current account deficit – capital flight, not trade (Buiter et al. 2011)



TARGET2: periphery central bank **liability** ↑, Bundesbank **claim** ↑

Takeaway: Inside a currency union a sudden stop shows up as official (Eurosystem) financing replacing private flows – a funding crisis, not a currency crisis; TARGET2 balances are a symptom, not the cause.

The Eurozone Policy Response (2010–2015)

What was done

Crisis management (short-run firefighting):

- **EFSF (2010) / ESM (2012):** conditional loans to Greece, Ireland, Portugal and Cyprus, and to Spain for bank recapitalisation; conditionality tied to fiscal consolidation and structural reform
- **ECB Securities Markets Programme (SMP, 2010):** sterilised secondary-market purchases of periphery sovereign bonds
- **ECB Outright Monetary Transactions (OMT, 2012):** Draghi's "whatever it takes" backstop – *unlimited*, conditional sovereign bond purchases

Structural reforms (medium run):

- **Banking union:** Single Supervisory Mechanism at the ECB (2014) and Single Resolution Mechanism (2016) – weakened the *bank–sovereign doom loop* by moving supervision and resolution above the national level
- **Fiscal Compact (2012):** treaty-level structural deficit limits with automatic correction mechanisms

The Eurozone Policy Response – What Remains Unresolved

Incomplete architecture

The euro area fixed the acute crisis but left root structural flaws largely in place.

- **No European Deposit Insurance Scheme (EDIS):** national deposit guarantee schemes remain, preserving the bank–sovereign link at the margin; the third pillar of banking union is still not agreed
- **No permanent fiscal union or common safe asset:** NextGenerationEU (2020) showed that large joint borrowing is possible, but it is temporary; there is no permanent euro-area safe asset that recycles surplus savings automatically (as US Treasuries do)

Takeaway: The response *stabilised* the crisis but did *not* resolve the tension between a monetary union, national fiscal policies and divergent competitiveness.

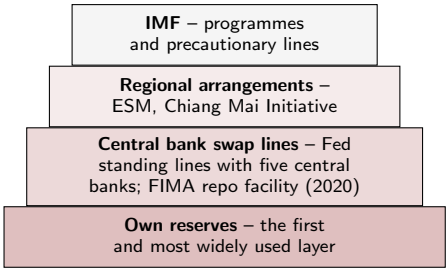
The Policy Toolkit for Capital Flows

Instrument	What it does	Examples and caveats
Exchange rate flexibility, monetary policy	First line of defence: the exchange rate absorbs shocks	Fear of floating where debt is in foreign currency
FX reserves and intervention	Self-insurance against stops; smooths disorderly moves	Asia's reserve build-up after 1997; costly to hold
Macroprudential measures (MPMs)	Limit FX lending, currency mismatches and leverage at home	Limits on FX mortgages after the CEE experience (FX lecture)
Capital flow management measures (CFMs)	Tax or limit inflows or outflows by residence	Chile's reserve requirement (1991–98); Brazil's tax on portfolio inflows (from 2009)

IMF Institutional View (2012, reviewed 2022)

CFMs should not substitute for needed macroeconomic adjustment. Since 2022, **pre-emptive** CFM/MPMs on debt inflows may be appropriate where stocks of foreign currency debt create systemic risk – targeted, temporary and transparent. The Integrated Policy Framework (Basu et al. 2020) studies how the four tools work together.

When Flows Stop: The Global Financial Safety Net



- The net is **uneven**: advanced economies with swap lines have ample cover; many emerging markets rely on their own reserves and the IMF
- Swap lines turn a dollar sudden stop into a manageable funding problem (2008, 2020; FX lecture, slide 81)
- IMF lending comes with conditionality and stigma – one reason EMs self-insure with reserves
- In the euro area, TARGET2 and the ECB's lending play the role of the safety net inside the union

The Czech Case: Capital Flows and the Koruna Commitment, 2013–2017

- **7 November 2013:** the ČNB commits to prevent the koruna from appreciating beyond CZK 27/EUR
- **From mid-2015:** inflows betting on the exit force heavy intervention – EUR 75 bn in total by March 2017
- Non-residents took 47% of government bond auctions in 2015–16 and 55% in the first half of 2017; short-term yields turned negative from August 2015. By 2025 non-residents held 25% of koruna government bonds (37% in 2019)
- **6 April 2017:** exit with immediate effect; the koruna appreciated gradually, without a gap

Why the exit was orderly

Positions were large but two-sided: exporters and investors sold the koruna as speculators bought it, so the central bank was not the only absorber (FX lecture: ČNB vs. SNB).

Takeaway: A credible one-way bet attracts large gross inflows into local bonds; reserves absorb them, and the exit tests liquidity and positioning.

Source: ČNB, Inflation Report II/2017, Box 1; Inflation Report III/2017, Box 2; ČNB, Zpráva o vývoji platební bilance 2025.

Scaling Up: From the Euro Area to Global Dollar Risk

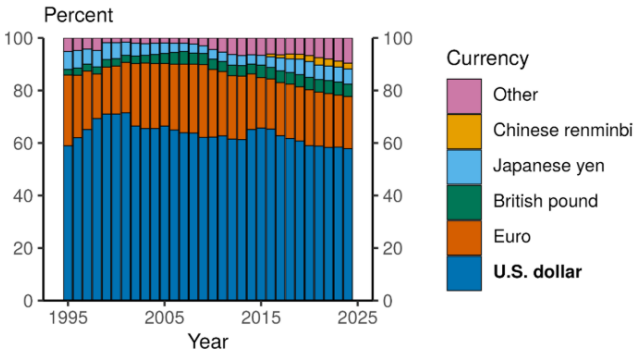
Euro-area lesson: sudden stops in gross flows can force painful internal adjustment when there is no exchange rate valve.

The same risk exists globally: the US runs persistent deficits financed by foreign gross inflows. What happens if those flows slow or reverse?

Reserve Currencies: The Basics

- **Definition:** foreign currencies held by central banks for intervention, international payments and as a store of value
- **Dominance:** driven by the issuer's economic size, stability and network effects (inertia)
- **Key characteristics:**
  - **Liquidity:** deep, accessible financial markets
  - **Stability:** low inflation and default risk
  - **Convertibility:** unrestricted capital flows
  - **Institutional strength:** rule of law, central bank independence

Global Foreign Exchange Reserves by Currency



Takeaway: The dollar's share has fallen from over 70% around 2000 to 56.7% (Q2 2026) – still more than the next three currencies combined; the renminbi remains small (2.1%).

Source: Federal Reserve Board (2025), "The international role of the U.S. dollar – 2025 edition", FEDS Notes; IMF COFER (Q2 2026).

Current Status of Reserve Currencies

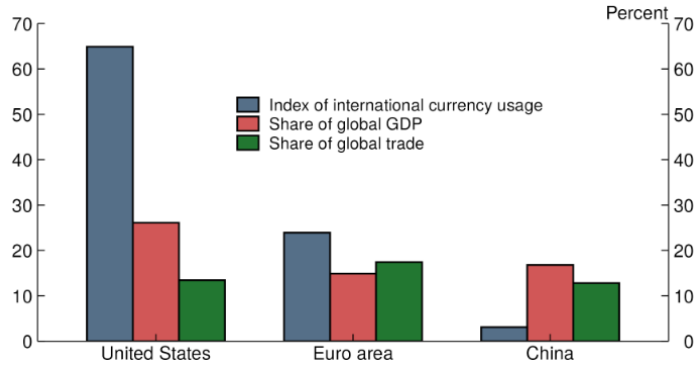
Currency	% (Q2 2026)
US dollar (USD)	56.70
Euro (EUR)	20.60
Japanese yen (JPY)	4.96
Chinese renminbi (CNY)	2.11
Other*	15.63

*GBP, CAD, AUD, CHF and others. Total reserves: USD 13.22 trillion.

- The dollar's share fell from 57.18% to 56.70% in Q2 2026, mainly because euro holdings rose
- Part of each quarter's change is valuation: when the dollar weakens, non-dollar reserves are worth more in dollars even if nobody sells dollars (IMF 2025)
- The renminbi is held back by capital controls and limited convertibility

Source: IMF blog, "Dollar's share of reserves held steady in second quarter when adjusted for FX moves", 2025.

International Role vs. Size of Economy (2024)



Takeaway: The dollar's international use (index about 65) far exceeds the US share of world GDP (about 26%) or trade (about 13%); the euro is roughly in proportion; the renminbi far below.

Source: Federal Reserve Board (2025), "The international role of the U.S. dollar – 2025 edition", FEDS Notes.

The US Dollar's Multifaceted Dominance

Why the dollar remains the world's currency

- **Reserves:** 56.7% of allocated global reserves (IMF COFER, Q2 2026)
- **FX trading:** on one side of about 89% of all FX trades (BIS Triennial Survey 2025)
- **Trade invoicing:** about 96% of trade invoicing in the Americas (Fed 2025)
- **Payments:** around half of SWIFT payments by value
- **The core support:**
  - The deepest, most liquid **Treasury market** (safe assets)
  - **Fed swap lines** (backstop liquidity in dollar stress)
  - A strong institutional framework (**rule of law**)

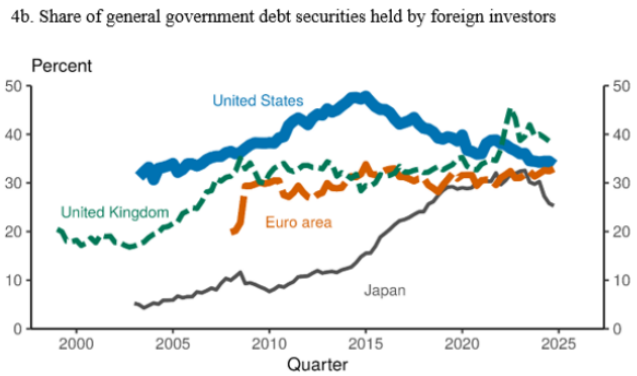
Geopolitics and Shifting Capital Flows

The consensus around the dollar as the ultimate safe haven is being tested by the use of reserves as a sanctions weapon (Russia, 2022) and by US fiscal dynamics.

- **Declining foreign share of US debt:** foreign investors hold about a third of US government debt securities, down from almost half in 2014 (next slide)
- **The rise of gold:** at market prices, foreign central banks' gold overtook their US Treasury holdings in 2025 – for the first time since 1996; gold was about 20% of global official reserves at end-2024, ahead of the euro (16%) (ECB 2025)
- **Fiscal dominance risks:** US net interest outlays ($881 bn in fiscal 2024, CBO) now exceed defence spending; markets increasingly ask whether Fed independence will hold
- **April 2025:** the dollar fell while US yields rose – unusual for a safe haven; FX lecture, slide 34, shows the hedging channel

Implication: global capital flows are fragmenting into blocs; deficit countries such as the US can no longer take foreign financing for granted.

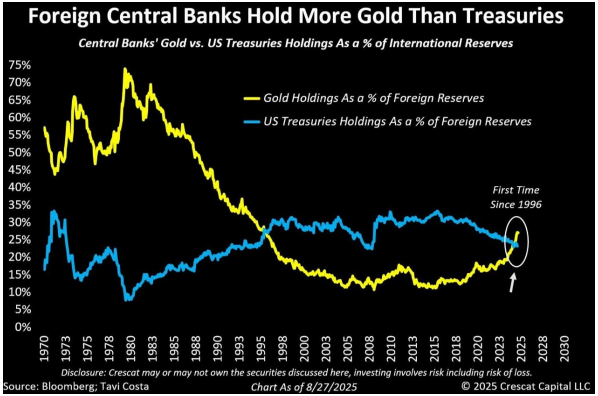
Government Debt Held by Foreign Investors



Takeaway: Foreigners hold about a third of US government debt, down from almost half in 2014 – similar to the euro area and the UK; the US now relies more on domestic buyers.

Source: Federal Reserve Board (2025), "The international role of the U.S. dollar – 2025 edition", FEDS Notes.

The Rise of Gold



Takeaway: At market prices, gold overtook Treasuries in foreign central banks' reserves in 2025 for the first time since 1996 – driven by purchases of over 1,000 tonnes a year in 2022–24 and by the gold price.

Source: Crescat Capital, data: Bloomberg; ECB (2025), The international role of the euro, June.

Conclusions (I): Vulnerabilities vs. Triggers

- **Accounting realities (the vulnerability):** current account deficits reflect saving–investment imbalances, $(S - I) + (T - G)$. They are a *medium- to long-term vulnerability gauge*, not an immediate crisis trigger
- **Look beyond the net:** financial crises are rarely caused by net trade imbalances. They are fuelled by the build-up of **gross capital flows** (bank lending, wholesale funding, portfolio flows), enabled by elastic credit creation
- **Sudden stops (the trigger):** crises are ignited by abrupt reversals in volatile gross flows, which create immediate liquidity and solvency problems and expose the underlying vulnerabilities

Conclusions (II): Policy, Distortions and Tail Risks

- **Policy:** monitor gross positions, currency mismatches and non-resident holdings; combine exchange rate flexibility, reserves, macroprudential measures and – where justified – CFMs; the global safety net is uneven
- **Asset market distortions:** persistent imbalances sustain asset booms and non-productive investment; large equity portfolio flows move valuations far from domestic fundamentals
- **Structural fragility and leverage:** dependence on foreign financing gives reserve holders leverage – for example by diversifying away from Treasuries
- **The geopolitical limit:** the dollar's exorbitant privilege is being tested; large fiscal deficits in a fragmenting world raise tail risks if foreign gross inflows stop

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